KUBERNETES:

- It is a container orchestration tool.
- Kubernetes is an open-source tool developed using GO programming language.
- Kubernetes is alias called as k8s
- k8s was developed by Google in 2014 it was made open source & donated to CNCF.

In Kubernetes we create cluster, in this cluster we will deploy our containerized application (docker images) in docker containers form.

This cluster is hosted by one node acting as the 'master' of the cluster, and other nodes as 'worker nodes' which do the actual 'containerization' using docker.

What Kubernetes can do ?

Deploy containers:	Runs your containers automatically across machines
Auto-scaling:	Adds more containers when traffic is high, removes when low
Auto-healing:	If a container crashes, it starts a new one automatically
Load balancing:	Spreads traffic evenly across all containers
Rolling updates:	Deploys new version of container without any downtime
Rollback:	If new version has a bug, goes back to old version automatically
Storage management:	Connects containers to cloud storage like AWS EBS
Networking:	Makes containers talk to each other and to the outside world
Security:	Controls who can access what using RBAC

Basic FAQs on Kubernetes

What is cluster?

cluster is a combination of k8s master node (control plane) + worker Nodes

What is Master Node?

- The master node is responsible for the management of worker nodes in k8s cluster.

  it has 4 main components kube-api server, kube scheduler, controller manager & etcd.

What is Worker Node?

- Worker nodes are the nodes where the container (application) actually running in kubernetes cluster.

  it has 3 main components kubelet (agent) , kube-proxy & container runtime (docker)

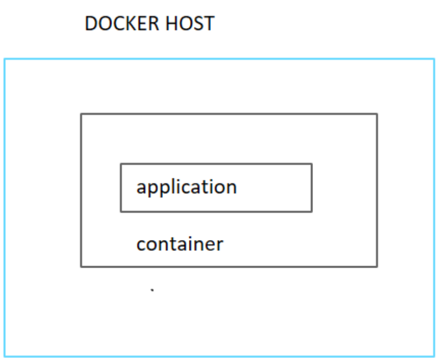
Can Kubernetes create docker images?

No, k8s cannot Create or modify the Docker images.

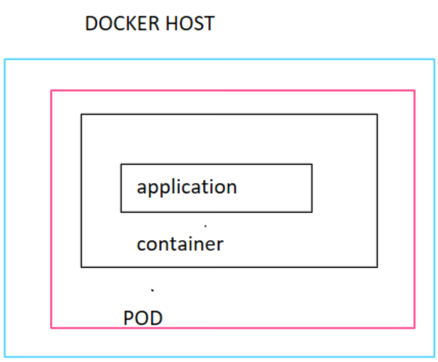
Can Kubernetes create containers?

Yes, **Kubernetes can create containers** but not **directly**.

Kubernetes creates an object called as **POD,** inside that pod containers will creates containers with the help of container runtimes like docker or containerd .



just docker



In Kubernetes

Is Kubernetes a container runtime like Docker?

No, Kubernetes is an **orchestration tool**, not a container runtime. It manages containers

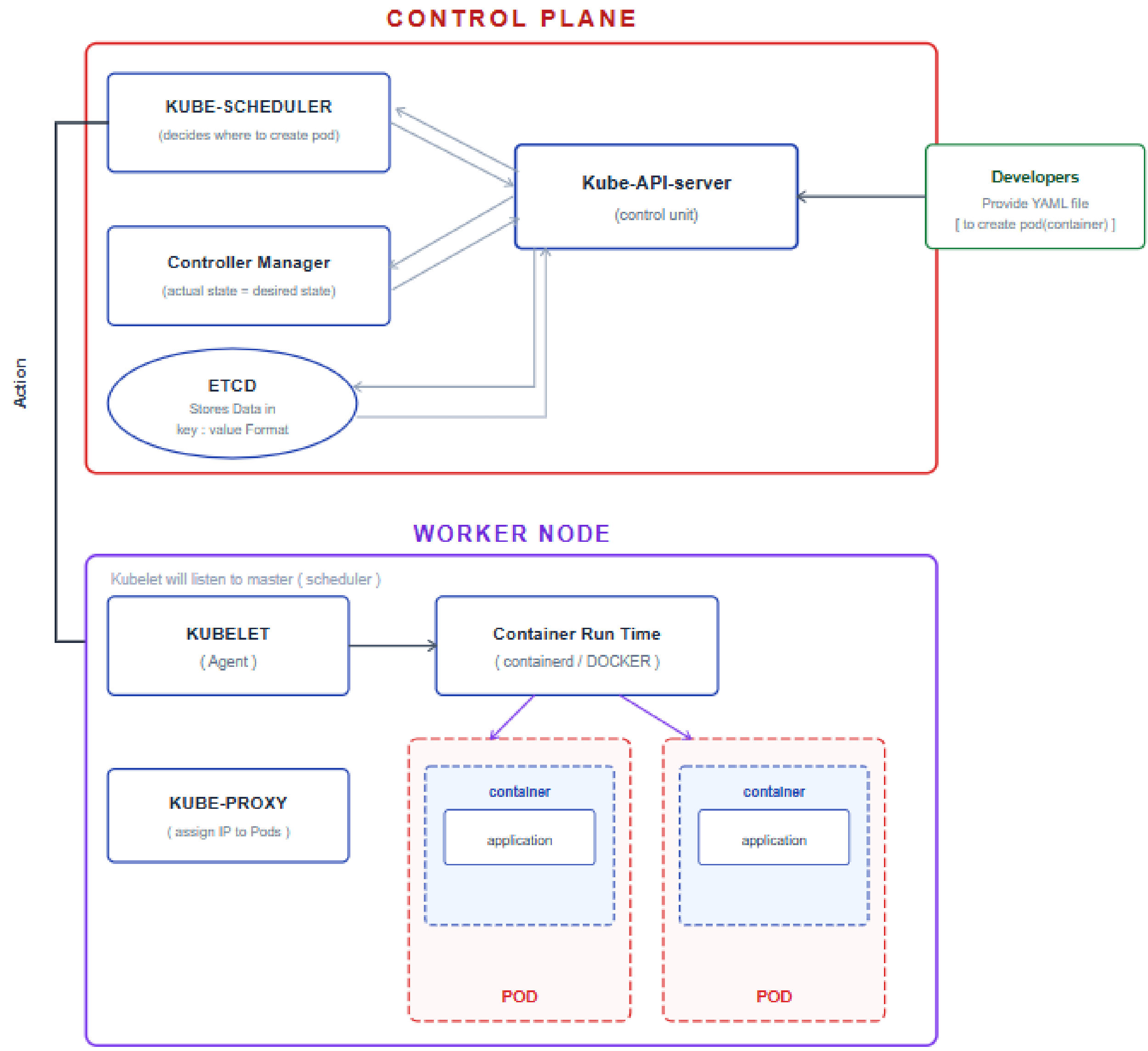
While containerization tools(or runtimes) like **Docker, containerD, and CRI-O** creates the containers

Can Kubernetes run without Docker?

Yes, Kubernetes supports **other container runtimes** like **containerD** and **CRI-O**, not just Docker.

Explain Kubernetes architecture?

Kubernetes cluster contains master node (control plane) with one or more worker nodes.



Master Node

- The master node is responsible for the management of Kubernetes cluster.

Master node contains 4 components.

1) kube-api server
2) kube scheduler
3) controller manager (actual state = desired state)
4) etcd

1) Kube-api server:

- ❖ Kube-api server is the main control center of a k8s cluster.

- It receives the requests (yaml file as input) from users (developers) for creation of k8s objects (like pods(containers))
- After checking the request (validating & storing details in etcd), it **forwards it to the Kube-Scheduler**.

2) Kube scheduler

- **Scheduler** gets input from the **Kube-API server**.
- It is responsible for **deciding where to run the pod** (on which worker node).
- It checks **available hardware resources** (like CPU, memory) on each node and picks the **best one**.
- It **doesn't create the pod**, just **decides where** it should be created.

3) Controller manager(controller):

- it is the brain behind orchestration process
- it **keeps watching the cluster** all the time. if any pods(container) **fails or goes down**, it will try to bring that up.
- It always makes sure that the **actual state** of the cluster **matches the desired state** (given in the YAML file).

4) etcd:

- ETCD is **like** a database to k8s cluster.
- Data is stored in key-value pair, format.
- all information about cluster will be stored here.
  - How many nodes are there
  - Which pods are running on which node
  - IP addresses of all pods
  - All configuration details

Worker Node:

Worker nodes are the nodes where the application(pods(containers)) actually running in kubernetes cluster.

Node contains 3 components

1) kubelet
2) container runtime
3) kube-proxy

1) Kubelet:

- **Kubelet** is an **agent** that runs on **every worker node** in the cluster.
- It **listens to the instructions from Master node** and follows instructions (like pod creation).
- Based on instructions from master, it **communicates with the container runtime** (like Docker or containerd) to **create containers** inside Pods.
- **Kubelet ensures** that the **Pod is running properly** on its node.
- It also sends **health/status updates** of the Pod to the Master.

2) Container runtimes (containerD)

- Container runtimes **receive instructions from Kubelet** to create containers.
- Container runtimes are responsible for pulling images and running containers inside Pods. (they will pull the images & creates container, exposes ports, volume mounts as defined in yaml file)

3) Kube-proxy

- Kube-proxy runs on every worker node.
- Kube-proxy is used for networking puprposes,
- like assigning ip address to each pods.
- Setting up communication between pods running on different nodes

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Full Flow — What happens when you deploy an app in Kubernetes

Developers write a YAML file

↓

kubectl sends it to kube-api server

↓

kube-api server validates and stores in etcd

↓

kube-scheduler decides which worker node to use

↓

kubelet on that worker node gets the instruction

↓

kubelet tells container runtime (containerd) to pull image and run container

↓

kube-proxy assigns IP to the pod

↓

Your container is running inside a Pod

What is pod?

Smallest deployable Object that kubernetes can create is pod.

Within the pod, we have our containers running & inside the container our application would be running.

Note on pods:

Pod → wrapper around container
Container → runs inside the Pod
Application → runs inside the Container

in 1 pod we can create 1-2 containers, but its best practice to run 1 container in a pod.

What is kubectl?

kubectl is a command-line tool used to interact with **Kubernetes**

All Kubernetes commands will always start with **kubectl.**

eg:

to list all pods ==> **`kubectl get pods`**

to list all nodes (master + worker nodes) ==> **`kubectl get nodes`**

Why did Kubernetes stop using Docker?

First, understand what Docker actually is — it is a full toolkit:

- A CLI to type commands
- A daemon (dockerd) to manage everything
- A registry to store images
- Networking and volume management
- And at the bottom — containerd, which actually runs the containers

When you use Docker, you need all of this together.

But Kubernetes does not need all of Docker features.

Kubernetes has its own:

- CLI — kubectl
- Scheduler and controller to manage containers
- Networking
- Image management

The only thing Kubernetes needs from Docker is "something that can run a container".

That something is containerd.

So when Kubernetes used Docker, the flow looked like this:

`kubectl → Kubernetes → dockerd → containerd → container runs`

Kubernetes was talking to dockerd just to reach containerd underneath. That is one extra unnecessary layer.

Now Kubernetes talks directly to containerd:

`kubectl → Kubernetes → containerd → container runs`

dockerd is removed from the picture entirely. Kubernetes goes straight to the part that actually does the work.

`Note:`

Since Kubernetes version **v1.24** Released in **April 2022 ,** kuberentes deprecated docker & started using only containerD

`Latest kuberentes version in v1.36`

How to get started with k8s?

Kubernetes can be installed using 2 or 3 ways

1. Self managed k8s clusters (kubeadm):

installing k8s software directly on servers by ourselves

we have to manage entire cluster

2. Managed Kubernetes Services (Cloud Providers):

To work with Kubernetes on AWS , we have a service **EKS** (Elastic Kubernetes Service)

Entire k8s cluster will be created by AWS cloud provider,

AWS will take care of maintaining k8s master,

we only need to add worker nodes to it.

Easiest & preferred way, but it is billed

3. Minikube:

Minikube is a tool that allows you to run a single-node Kubernetes cluster locally on your machine.

Minikube is designed to enable developers to develop and test applications locally before deploying them to a full-scale Kubernetes cluster.

generally used by developers

4. **Online Kubernetes Platforms**:

Best for Quick experimentation without local setup

No installation of any softwares, accessible from browser for free

Cons: Temporary environments, limited resources

- Killercoda
- Kubernetes Playground
- Play with Kubernetes

YAML:

- **yaml** is file format Human-readable data serialization format
- yaml is space indented (equal spaces to be given)
- in yaml data would be in key value format (Dictionary format)
- always whenever we write yaml file it will start with --- & ends with ...
- # Comments start with a hash symbol
- yaml files extension can be .yml (or) .yaml

Example yaml file:

```yaml
---
# Root/Parent Element
my-yaml-file:
  # Child Elements (key1, key2, key3)
  - key1: value1
  - key2:
      subkey1: value1  # Sub-element of key2
      subkey2: value2  # Sub-element of key2
  - key3:
      subkey:
        sub-subkey1: value  # Nested sub-element
        sub-subkey2: value  # Nested sub-element

# Another top-level key
anotherkey: another_value
...
```

Structure Explanation:

1. **Root Element**: my-yaml-file (parent to all keys)
2. **Child Elements**:
   - key1: Simple key-value pair
   - key2: Parent with two subkeys
   - key3: Parent with nested sub-subkeys
3. **Standalone Top-Level Key (**anotherkey**)**
   - Exists at the same level as my-yaml-file
   - Not part of the my-yaml-file structure
   - Shows multiple root-level elements can coexist

Important Notes:

- Proper indentation (2 spaces per level)
- Lists are marked with - (hyphen + space)
- Child elements are indented under parents

Examples of list & key value pairs

```yaml
---
fruits:           # Parent (root element)
  - Apple         # │
  - Banana        # │
  - Mango         # ├─ Children (all equal siblings)
  - Grapes        # │
  - Watermelon    # │
...
```

Here **Parent**: fruits (root-level key) & it has **5 Children**: Apple, Banana, Mango, Grapes, Watermelon (list items)

```yaml
---
# Indian Cricket Team Players by Roles
indian_cricket_team:    # parent: root
  - coach: dravid       # child of indian_cricket_team
  - captain: rohit      # child of indian_cricket_team
  - wicketkeeper: rahul # child of indian_cricket_team

  - batsmen:            # child of indian_cricket_team
      - virat           # child of batsmen, sibling of rohit and rahul
      - rohit           # child of batsmen, sibling of virat and rahul
      - rahul           # child of batsmen, sibling of virat and rohit

  - bowlers:            # child of indian_cricket_team
      - bumrah          # child of bowlers, sibling of shami and siraj
      - shami           # child of bowlers, sibling of bumrah and siraj
      - siraj           # child of bowlers, sibling of bumrah and shami

  - all_rounders:       # child of indian_cricket_team
      - hardik          # child of all_rounders (only child)
...
```

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```yaml
---
#Example3: whatever we have learnt so far in devops course in yaml format
DEVOPS:
 os: linux
 script: shellscripting
 scm:
  - git
  - github
 cicd:
  - jenkins
  - gitlab
 conatiners:
   - docker
   - containerD
   - crio
 orchetarstartion: k8s
...
```

to validate yaml file syntax goto **www.yamllint.com** paste & check for file format / indentation check.

it shows our yaml is valid or not.

YAML files in kubernetes:

In K8S These yaml files are also called as manifest files (or) definition file.

As a devops engineers, we need to create a yaml files (.yml) file

What this yaml file contains?

1) how many containers needs to be create in a pod.

2) what name we need to give for pod & container.

3) what image we need to use to create container.

etc...

All the above information will be available in k8s yaml file.

This yaml file / definition file / manifest file should be provided to kubernetes master.